



FACULTY OF INFORMATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE

ICS 1204: INTEGRAL CALCULUS

Date: March 1, 2017

Time: 2 hours

INSTRUCTIONS: Answer QUESTION ONE and any other TWO questions.

QUESTION ONE (30 MARKS)

- (a) Find the area under the parabola $y = x^2 - 2x + 2$, above the x-axis and between $x=0$ and $x=1$. [4 Marks]
- (b) Solve the following differential equations $\frac{dy}{dx} = 2x(1 + y)$. Hence find a particular solution if $x = 0$ and $y = 2$. [6 Marks]
- (c) Evaluate the following definite integrals
- (i) $\int_0^1 \frac{1}{x^2} + (x - 1)^2 dx$ [3 Marks]
- (ii) $\int_0^2 xe^x dx$ [3 Marks]
- (iii) $\int_0^{\pi/2} \sin^2 x \cos x dx$ [3 Marks]
- (iv) $\int_1^4 \frac{(x+1)(x-2)}{\sqrt{x}} dx$ [3 Marks]
- (d) Find the Taylor polynomials of degree 3 for $f(x) = \frac{1}{1-x}$, centered at $x = 0$. [4 Marks]
- (e) Given that the half-life T of radium is 1690 years, how much will remain of one gram of radium after 1000 years? [4 Marks]

QUESTION TWO (20 MARKS)

- (a) Use an appropriate method to evaluate the following integrals:
- (i) $\int x^2 \ln x dx$ [4 Marks]

(ii) $\int \tan^{-1} x \, dx$ [4 Marks]

(b) Determine the volume of the solid of revolution formed by revolving the area enclosed by the parabola $y = 4x^2$ and the lines $x = 0$ and $y = 16$. [5 Marks]

(c) Find the area $\mathbb{A}$ of the region $\mathbb{R}$ under the line $y = \frac{1}{2}x + 2$, above the parabola $y = x^2$, and between the y -axis and $x = 1$. [7 Marks]

QUESTION THREE (20 MARKS)

(a) Solve the following differential equation:

$(x + 2) \frac{dy}{dx} = (2x^2 + 4x + 1)(y - 3)$, given that $y(0) = 7$. [5 Marks]

(b) Use integration by parts to verify the specified formula.

$$\int e^x \cos x \, dx = \frac{1}{2} e^x (\sin x + \cos x) + C$$

[5 Marks]

(c) Find:

(i) $\int \frac{x^4 + 2x^3 - 4x^2 + x - 3}{x^2 - x - 2} \, dx$ [5 Marks]

(ii) $\int \frac{2x^2 + 5x - 1}{x(x^2 - 1)} \, dx$ [5 Marks]

QUESTION FOUR (20 MARKS)

(a) Assume that the number of bacteria in a culture grows exponentially with a growth constant of 0.02, time is measured in hours.

(i) How many bacteria will be present after 1 hour if there are initially 1000? [3 Marks]

(ii) Given the same initial 1000 bacteria, in how many hours will there be 100,000 bacteria? [3 Marks]

(b) Determine the area of the region bounded by the given curves: $y = x^2$, $y = 0$, $x = 2$, $x = 5$. [5 Marks]

(c) Evaluate:

(i) $\int_{-3}^3 \sqrt{9 - x^2} \, dx$ [5 Marks]

(ii) $\int \frac{dx}{1 + e^x}$ [4 Marks]

QUESTION FIVE (20 MARKS)

- (a) Resolve the following rationals into partial fractions

$$\frac{x^3 - 6x + 3}{(x - 2)^3}.$$

Hence evaluate $\int \frac{x^3 - 6x + 3}{(x - 2)^3} dx$

[8 Marks]

- (b) Use an appropriate method to evaluate the following integrals:

(i) $\int \sin\left(\frac{x}{2}\right) dx$

[3 Marks]

(ii) $\int \frac{dx}{1 + \cos x} dx$ [Hint: Rationalize the denominator]

[5 Marks]

(iii) $\int \frac{4x^7}{3x^8 - 2} dx$

[4 Marks]

END OF PAPER